```python
import pandas as pd

# Load the dataset
diabetes_df = pd.read_csv("diabetes.csv")

# Display the first few rows
print(diabetes_df.head())
```

[1] ✓ 17.9s

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
```

```python
# Dataset structure and summary statistics
print(diabetes_df.info())
print(diabetes_df.describe())

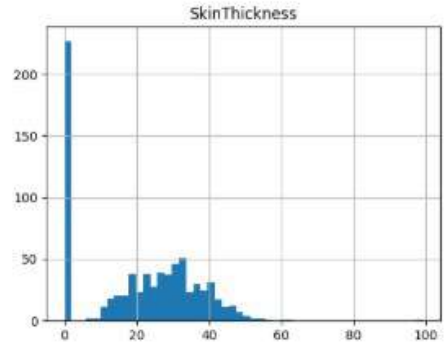
# Visualize the distribution of each feature
import matplotlib.pyplot as plt
import seaborn as sns

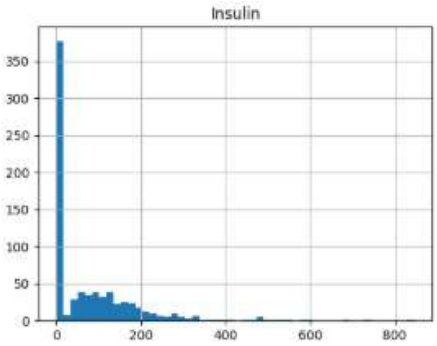
diabetes_df.hist(bins=50, figsize=(20, 15))
plt.show()

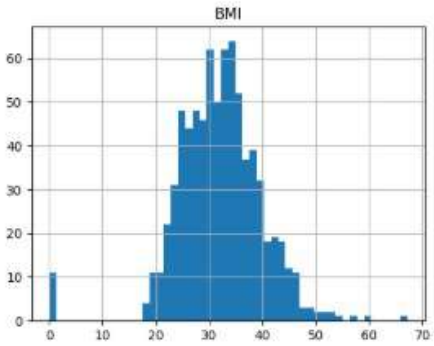
# Pairplot to visualize relationships
sns.pairplot(diabetes_df)
plt.show()
```

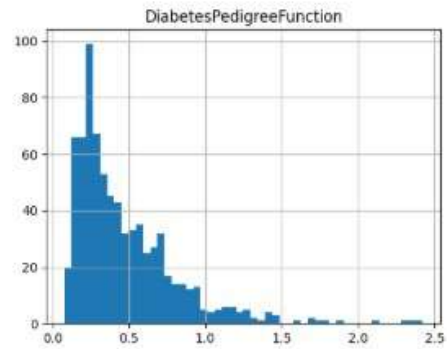
[2] ✓ 38.4s

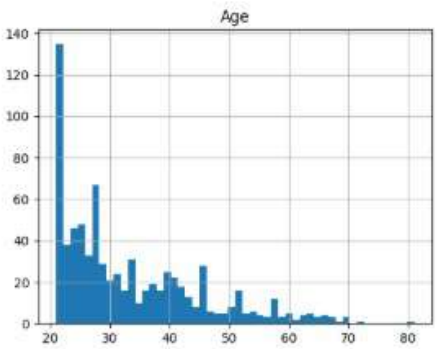
```
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #  Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0  Pregnancies               768 non-null    int64
 1  Glucose                   768 non-null    int64
 2  BloodPressure             768 non-null    int64
 3  SkinThickness             768 non-null    int64
 4  Insulin                   768 non-null    int64
 5  BMI                       768 non-null    float64
 6  DiabetesPedigreeFunction  768 non-null    float64
 7  Age                       768 non-null    int64
 8  Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
```

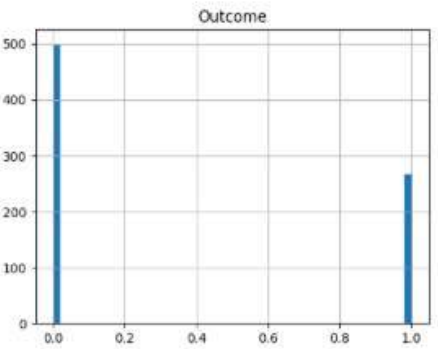

SkinThickness
0
50
100
150
200
0
20
40
60
80
100

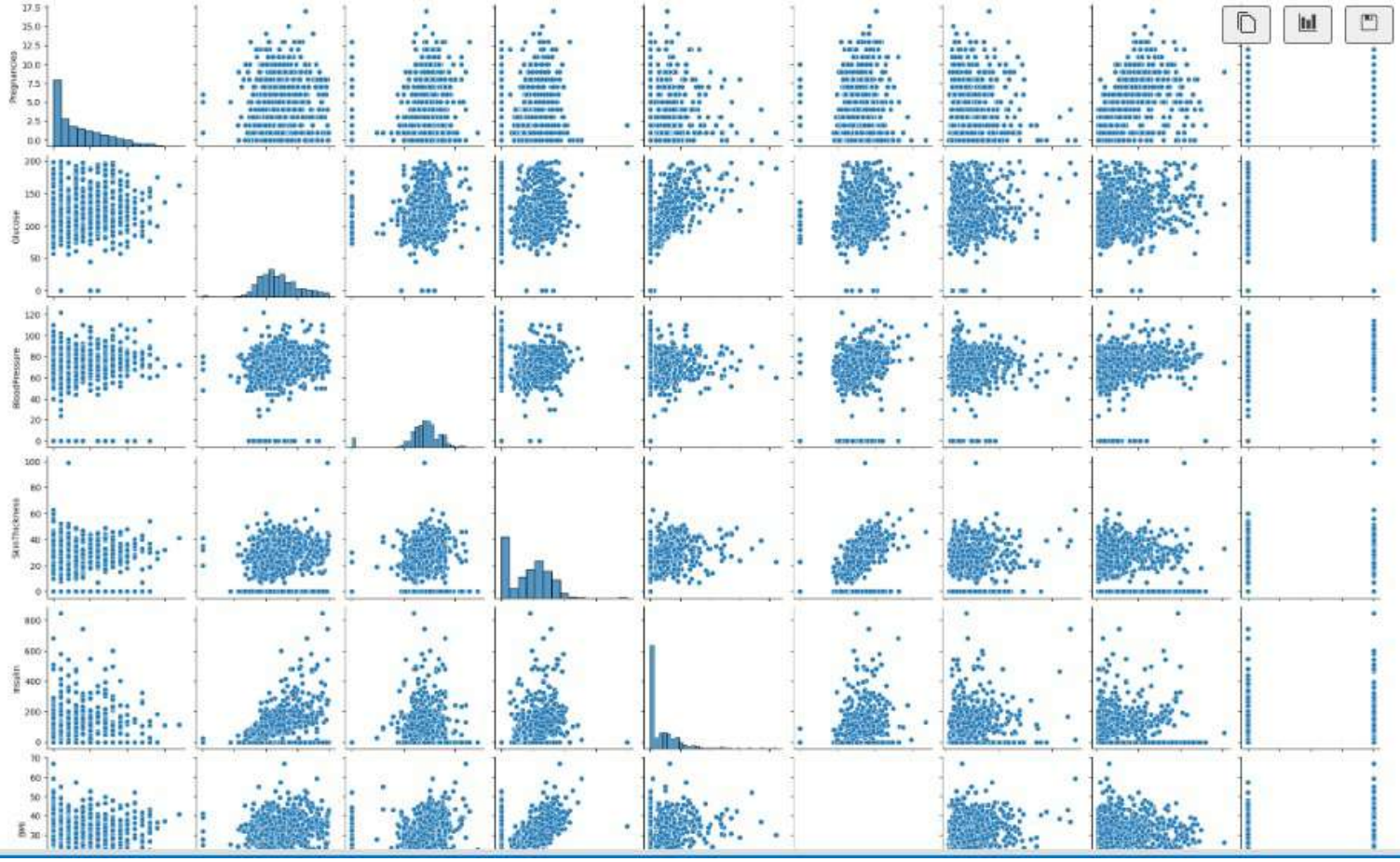

Insulin
0
50
100
150
200
250
300
350
0
200
400
600
800


BMI
0
10
20
30
40
50
60
0
10
20
30
40
50
60
70


DiabetesPedigreeFunction
0
20
40
60
80
100
0.0
0.5
1.0
1.5
2.0
2.5


Age
0
20
40
60
80
100
120
140
20
30
40
50
60
70
80

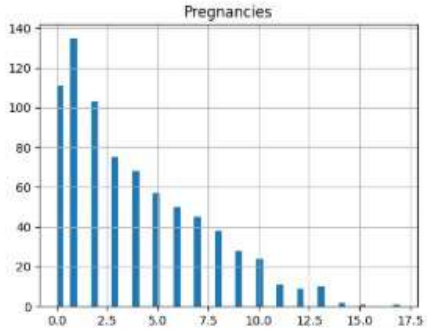

Outcome
0
100
200
300
400
500
0.0
0.2
0.4
0.6
0.8
1.0

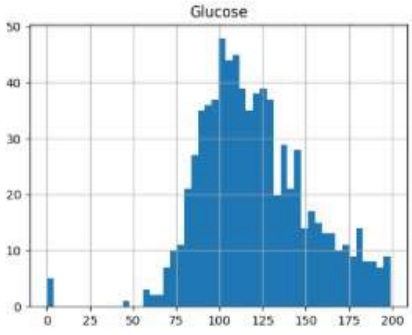

Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI

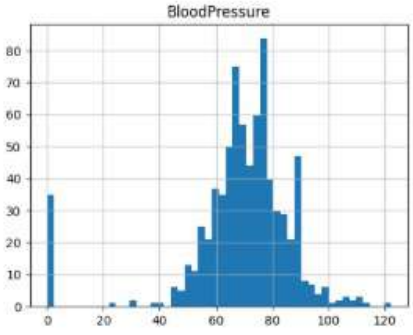
None

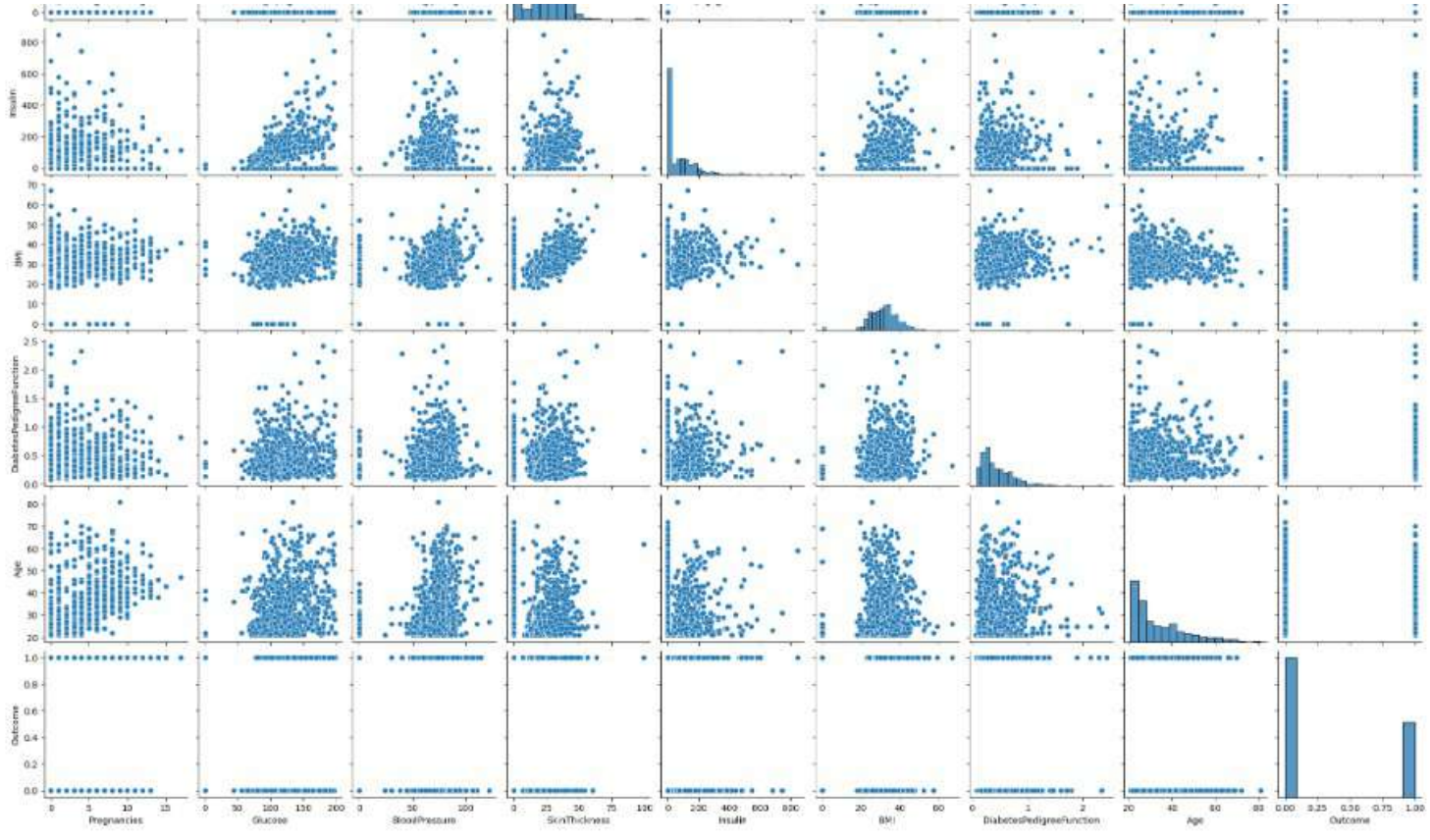
| | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin \ |
|---|---|---|---|---|---|
| count | 768.000000 | 768.000000 | 768.000000 | 768.000000 | 768.000000 |
| mean | 3.845052 | 120.894531 | 69.105469 | 20.536458 | 79.799479 |
| std | 3.369578 | 31.972618 | 19.355807 | 15.952218 | 115.244002 |
| min | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 |
| 25% | 1.000000 | 99.000000 | 62.000000 | 0.000000 | 0.000000 |
| 50% | 3.000000 | 117.000000 | 72.000000 | 23.000000 | 30.500000 |
| 75% | 6.000000 | 140.250000 | 80.000000 | 32.000000 | 127.250000 |
| ... | | | | | |
| 25% | 27.300000 | | 0.243750 | 24.000000 | 0.000000 |
| 50% | 32.000000 | | 0.372500 | 29.000000 | 0.000000 |
| 75% | 36.600000 | | 0.626250 | 41.000000 | 1.000000 |
| max | 67.100000 | | 2.420000 | 81.000000 | 1.000000 |

*Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...*




Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome